WEEK 6 OVERVIEW

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CEVE 543

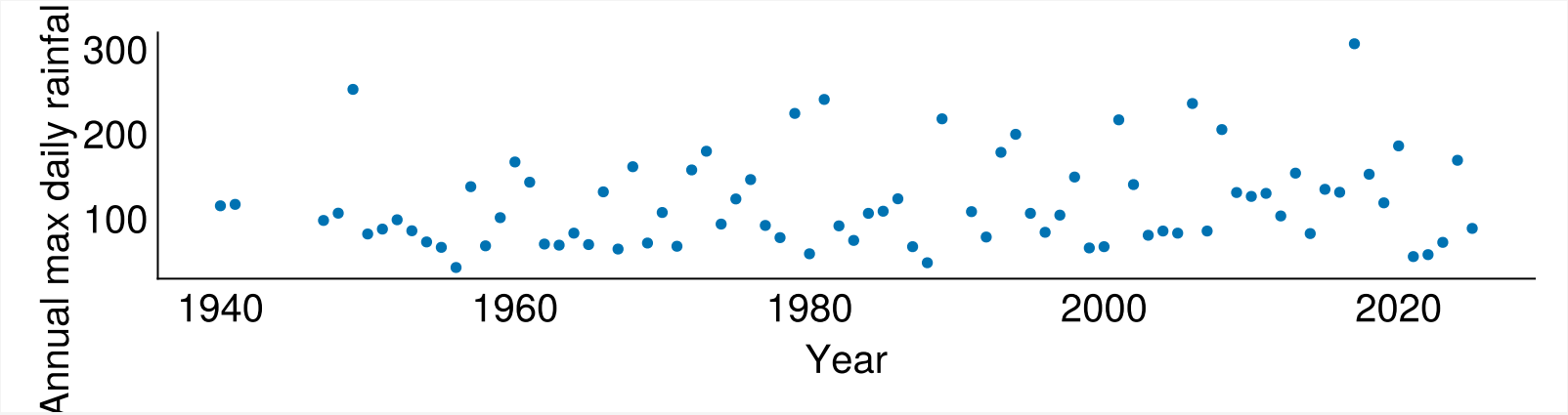
2026-09-28



THIS WEEK

- **Monday:** nonstationarity, trend tests, and covariates
- **Wednesday:** Smith et al. (2026); bring your two questions
- **Friday:** lab 6, fitting nonstationary extreme value models

ANNUAL MAXIMUM RAINFALL AT HOUSTON HOBBY

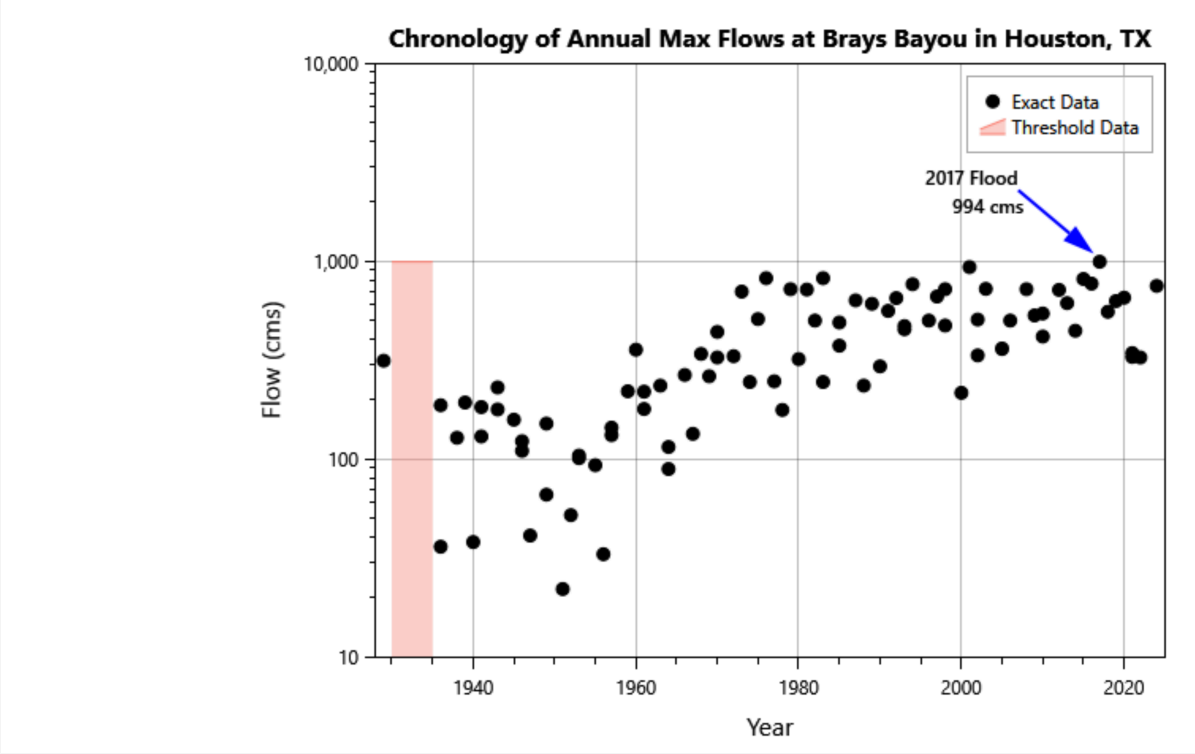


Houston Hobby Airport (GHCN-Daily USW00012918), 80 years with at least 350 days of data.

Think

Last week you subtracted mean sea level from the Galveston record before fitting. What would you subtract here?

A TREND MEANS THE YEARS ARE NOT IDENTICALLY DISTRIBUTED



Brays Bayou at Houston, annual peak flow, 1929-2024 (Smith et al., 2026, Fig. 2, CC BY 4.0).

SUBTRACTING A TREND IS NOT ENOUGH

- Galveston: sea level adds to surge, so subtracting it shifts the distribution and keeps its shape.
- Brays Bayou: after about 1968 the typical flood rose and the spread, on a log scale, narrowed (Smith et al., 2026, pp. 14, 24).
- No single number subtracted from each year removes both changes.

STATIONARY GIVEN A COVARIATE

Let each year's distribution depend on a covariate x_t, such as sea level or CO2:

$$Y_t \mid x_t \sim \text{GEV}(\mu(x_t), \sigma(x_t), \xi)$$

- μ, σ, and ξ are the location, scale, and shape from week 4.
- Given the covariate, the years are independent: **conditional independence**.
- Subtracting mean sea level at Galveston was the special case $\mu(x_t) = \mu_0 + x_t$, with σ and ξ fixed.
- Whether a record is "stationary" depends on what you condition on.